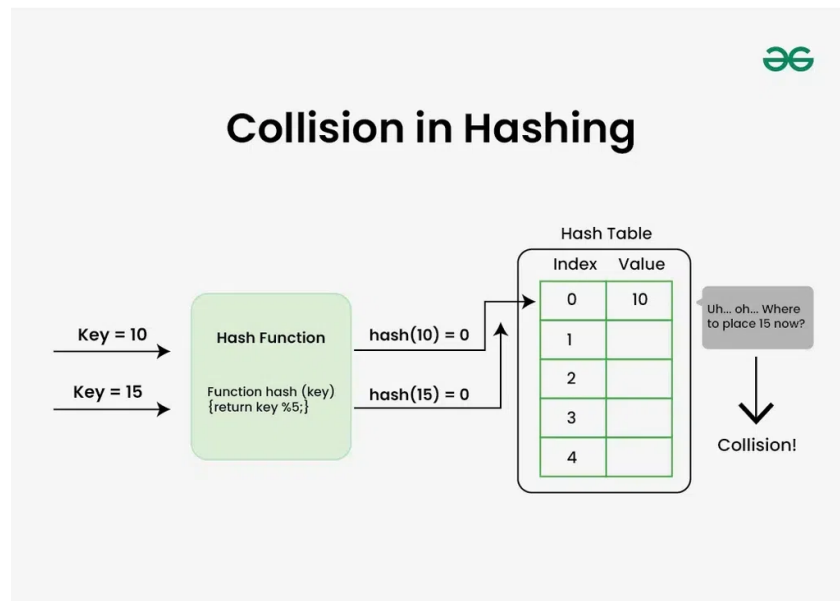


Collision Resolution Techniques

Last Updated : 10 Mar, 2025



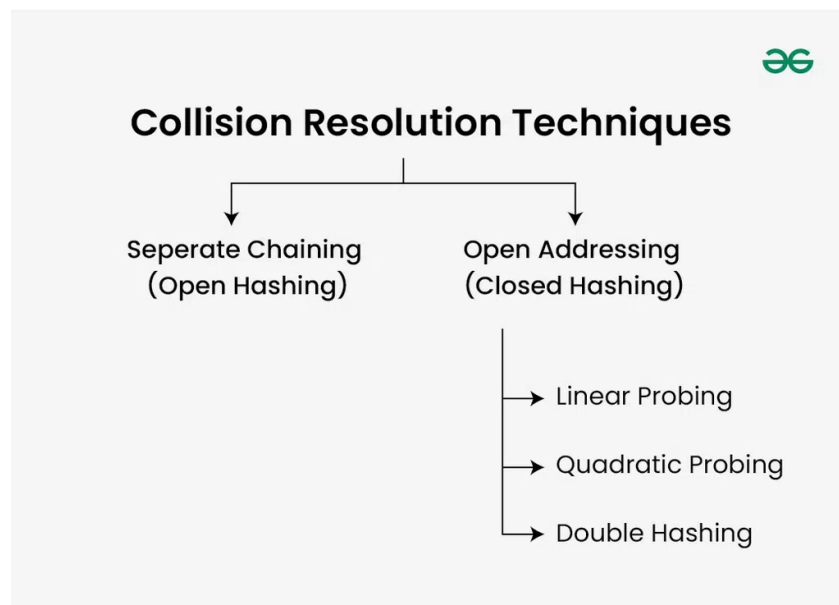
In **Hashing**, hash functions were used to generate hash values. The hash value is used to create an index for the keys in the hash table. The hash function may return the same hash value for two or more keys. When two or more keys have the same hash value, a **collision** happens. To handle this collision, we use **Collision Resolution Techniques**.



Collision Resolution Techniques

There are mainly two methods to handle collision:

1. Separate Chaining
2. Open Addressing



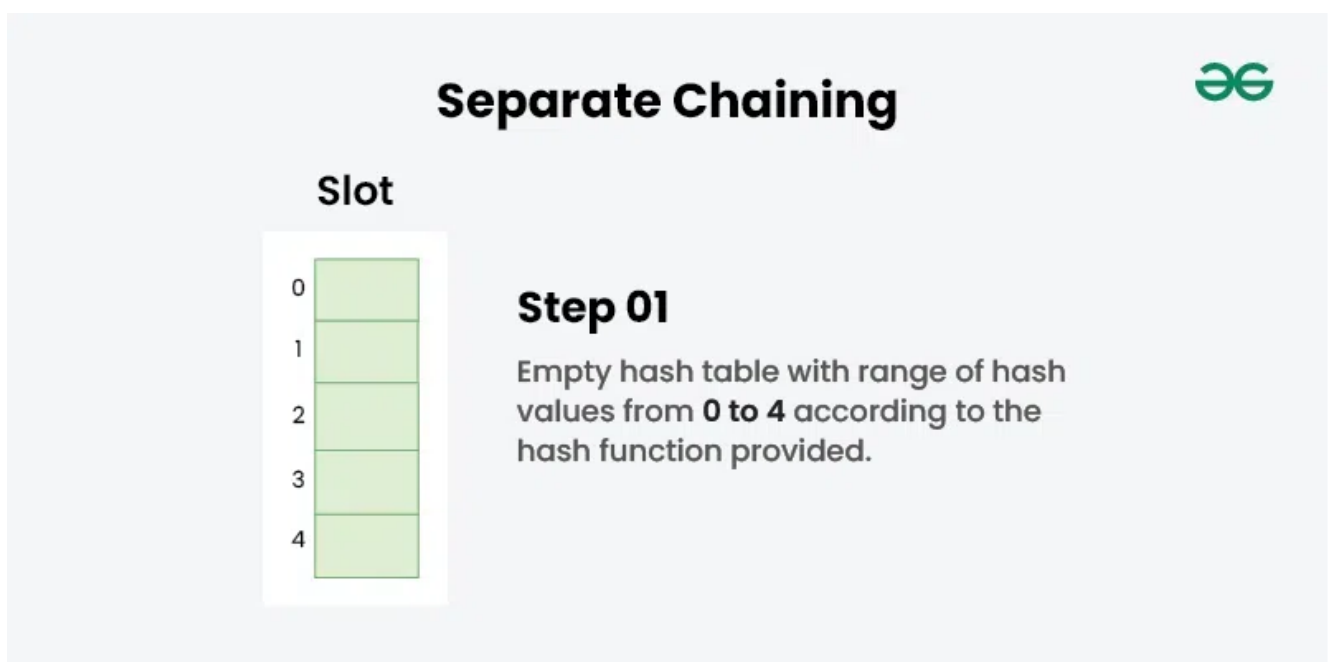
1) Separate Chaining

The idea behind [Separate Chaining](#) is to make each cell of the hash table point to a linked list of records that have the same hash function value. Chaining is simple but requires additional memory outside the table.

Example: We have given a hash function and we have to insert some elements in the hash table using a separate chaining method for collision resolution technique.

*Hash function = $\text{key} \% 5$,
Elements = 12, 15, 22, 25 and 37.*

Let's see step by step approach to how to solve the above problem:



Separate Chaining



Slot

0	
1	
2	12
3	
4	

Step 02

The first key to be inserted is **12** which is mapped to slot 2 ($12\%5=2$).

Separate Chaining



Slot

0	
1	
2	12
3	
4	

→ 22

Step 03

The next key is **22** which is mapped to slot 2 ($22\%5=2$) but slot 2 is already occupied by key 12. Separate chaining will handle collision by creating a linked list to slot 2.

Separate Chaining



Slot

0	15
1	
2	12
3	
4	

→ 22

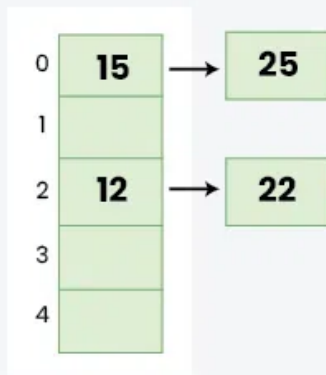
Step 04

The next key is **15** which is mapped to slot 0 ($15\%5=0$).

Separate Chaining



Slot



Step 05

The next key is 25 which is mapped to slot 0 ($25\%5=0$). But slot 0 is already occupied by key 25. Again, Separate chaining will handle collision by creating a linked list to slot 2.

Please [You Own Hash Table with Chaining](#) for implementation of this technique

2) Open Addressing

In [open addressing](#), all elements are stored in the hash table itself. Each table entry contains either a record or NIL. When searching for an element, we examine the table slots one by one until the desired element is found or it is clear that the element is not in the table.

2.a) Linear Probing

In linear probing, the hash table is searched sequentially that starts from the original location of the hash. If in case the location that we get is already occupied, then we check for the next location.

Algorithm:

1. Calculate the hash key. i.e. **$key = data \% size$**
2. Check, if **$hashTable[key]$** is empty
 - store the value directly by **$hashTable[key] = data$**
3. If the hash index already has some value then
 - check for next index using **$key = (key+1) \% size$**
4. Check, if the next index is available **$hashTable[key]$** then store the value. Otherwise try for next index.
5. Do the above process till we find the space.

Example: Let us consider a simple hash function as “key mod 5” and a sequence of keys that are to be inserted are 50, 70, 76, 85, 93.



Linear Probing (Open Addressing)



Slot

0	
1	
2	
3	
4	

Step 01

Empty hash table with range of hash values from **0 to 4** according to the hash function provided.

Linear Probing (Open Addressing)



Slot

0	50
1	
2	
3	
4	

Step 02

The first key to be inserted is **50** which is mapped to **slot 0** ($50\%5=0$)

Linear Probing (Open Addressing)



Slot

0	50
1	70
2	
3	
4	

Step 03

The next key is **70** which is mapped to **slot 0** ($70\%5=0$) but **50** is already at **slot 0** so, search for the next empty slot and insert it.

Linear Probing (Open Addressing)



Slot

0	50
1	70
2	76
3	
4	

Step 04

The next key is **76** which is mapped to **slot 1** ($76\%5=1$) but **70** is already at **slot 1** so, search for the next empty slot and insert it.

Linear Probing (Open Addressing)



Slot

0	50
1	70
2	76
3	85
4	

Step 05

The next key is **85** which is mapped to **slot 0** ($85\%5=0$), but **50** is already at **slot number 0** so, search for the next empty slot and insert it. So insert it into **slot number 3**.

Linear Probing (Open Addressing)



Slot

0	50
1	70
2	76
3	85
4	93

Step 06

The next key is **93** which is mapped to **slot 3** ($93\%5=3$), but **85** is already at **slot 3** so, search for the next empty slot and insert it. So insert it into **slot number 4**.

Please refer [Your Own Hash Table with Linear Probing in Open Addressing](#) for implementation details.

2.b) Quadratic Probing

Quadratic probing is an open addressing scheme in computer programming for resolving hash collisions in hash tables. Quadratic probing operates by taking the original hash index and adding successive values of an arbitrary quadratic polynomial until an open slot is found.

An example sequence using quadratic probing is:

$$H + 1^2, H + 2^2, H + 3^2, H + 4^2, \dots, H + k^2$$

This method is also known as the mid-square method because in this method we look for i^2 -th probe (slot) in i -th iteration and the value of $i = 0, 1, \dots, n - 1$. We always start from the original hash location. If only the location is occupied then we check the other slots.

Let $\text{hash}(x)$ be the slot index computed using the hash function and n be the size of the hash table.

*If the slot $\text{hash}(x) \% n$ is full, then we try $(\text{hash}(x) + 1^2) \% n$.
If $(\text{hash}(x) + 1^2) \% n$ is also full, then we try $(\text{hash}(x) + 2^2) \% n$.
If $(\text{hash}(x) + 2^2) \% n$ is also full, then we try $(\text{hash}(x) + 3^2) \% n$.
This process will be repeated for all the values of i until an empty slot is found*

Example: Let us consider table Size = 7, hash function as $\text{Hash}(x) = x \% 7$ and collision resolution strategy to be $f(i) = i^2$. Insert = 22, 30, and 50

Quadratic Probing (Open Addressing)



Slot

0	
1	
2	
3	
4	
5	
6	

Step 01

Empty hash table with range of hash values from 0 to 6 according to the hash function provided.

Quadratic Probing (Open Addressing)



Slot

0	
1	22
2	
3	
4	
5	
6	

Step 02

The first key to be inserted is 22 which is mapped to slot 1 ($22 \% 7 = 1$)

Quadratic Probing (Open Addressing)



Slot

0	
1	22
2	30
3	
4	
5	
6	

Step 03

The next key is 30 which is mapped to slot 2 ($30 \% 7 = 2$)

Quadratic Probing (Open Addressing)



Slot	
0	
1	22 ← $1+0$
2	30 ← $1+1^2$
3	
4	
5	50 ← $1+2^2$
6	

Step 04

The next key is 50 which is mapped to slot 1 ($50\%7=1$) but slot 1 is already occupied. So, we will search slot $1+1^2$, i.e. $1+1 = 2$. Again slot 2 is occupied, so we will search cell $1+2^2$, i.e. $1+4 = 5$,

Please refer [Your Own Hash Table with Quadratic Probing in Open Addressing](#) for implementation.

2.c) Double Hashing

Double hashing is a collision resolving technique in [Open Addressed](#) Hash tables. Double hashing make use of two hash function,

- The first hash function is **$h_1(k)$** which takes the key and gives out a location on the hash table. But if the new location is not occupied or empty then we can easily place our key.
- But in case the location is occupied (collision) we will use secondary hash-function **$h_2(k)$** in combination with the first hash-function **$h_1(k)$** to find the new location on the hash table.

This combination of hash functions is of the form

$$h(k, i) = (h_1(k) + i * h_2(k)) \% n$$

where

- i is a non-negative integer that indicates a collision number,

- k = element/key which is being hashed
- n = hash table size.

Complexity of the Double hashing algorithm:

Time complexity: $O(n)$

Example: Insert the keys 27, 43, 692, 72 into the Hash Table of size 7. where first hash-function is $h_1(k) = k \bmod 7$ and second hash-function is $h_2(k) = 1 + (k \bmod 5)$

Double Hashing (Open Addressing)



Slot

0	
1	
2	
3	
4	
5	
6	

Step 01

Empty hash table with range of hash values from 0 to 6 according to the hash function provided.

Double Hashing (Open Addressing)



Slot

0	
1	
2	
3	
4	
5	
6	27

Step 02

The first key to be inserted is 27 which is mapped to slot 6 ($27 \bmod 7 = 6$).

Double Hashing (Open Addressing)



Slot

0	
1	43
2	
3	
4	
5	
6	27

Step 03

The next key is **43** which is mapped to slot 1 ($43 \% 7 = 1$).

Double Hashing (Open Addressing)



Slot

0	
1	43
2	692
3	
4	
5	
6	27

Step 04

The next key is **692** which is mapped to slot 6 ($692 \% 7 = 6$), but location 6 is already occupied.

Using double hashing,

$$h_{new} = [h_1(692) + i * (h_2(692))] \% 7$$

$$= [6 + 1 * (1 + 692 \% 5)] \% 7$$

$$= 9 \% 7$$

$$= 2$$

Now, as 2 is an empty slot, so we can insert 692 into 2nd slot.

Double Hashing (Open Addressing)



Slot

0	
1	43
2	692
3	
4	
5	72
6	27

Step 05

The next key is **72** which is mapped to slot 2 ($72 \% 7 = 2$), but location 2 is already occupied.

Using double hashing,

$$h_{new} = [h_1(72) + i * (h_2(72))] \% 7$$

$$= [2 + 1 * (1 + 72 \% 5)] \% 7$$

$$= 5 \% 7$$

$$= 5,$$

Now, as 5 is an empty slot, so we can insert 72 into 5th slot.

Please refer [Double Hashing](#) for Implementation.

 Comment

More info 

Advertise with us

Next Article 

Collision Resolution Techniques

Similar Reads

First collision point of two series

Given five numbers a , b , c , d and n (where $a, b, c, d, n > 0$). These values represent n terms of two series. The two series formed by these four numbers are $b, b+a, b+2a, \dots, b+(n-1)a$ and $d, d+c, d+2c, \dots, d+(n-1)c$...

🕒 7 min read

Types of Collisions

Collisions are an everyday phenomenon that occurs when two or more objects come into contact with each other, often resulting in a transfer of energy. They are classified into different types based on how...

🕒 10 min read

Inelastic Collision

Inelastic Collision is a type of collision where momentum is conserved, but kinetic energy is not. In such collisions, the colliding objects stick together, and some kinetic energy is transformed into other forms li...

🕒 9 min read

Techniques of Cyber Forensics

Digital Forensics, or Computer Forensics, is the applied scientific methodology and procedures in identifying, collecting, protecting, examining, presenting, and legally preserving digital evidence in any...

🕒 5 min read

Elastic vs Inelastic Collision in one Dimension

During the game, you may have witnessed two billiard balls collide with each other. A collision is the violent coming together of two distinct bodies. What happens when two objects collide? Can we identif...

🕒 10 min read

Elastic Collision Formula

Elastic Collision is the collision in which the kinetic energy of the system is conserved. A collision occurs when two or more object comes in direct contact with each other. the collision can be of two types that...

🕒 9 min read

Two Pointers Technique

Two pointers is really an easy and effective technique that is typically used for Two Sum in Sorted Arrays, Closest Two Sum, Three Sum, Four Sum, Trapping Rain Water and many other popular interview...

🕒 11 min read

Open Addressing Collision Handling technique in Hashing

Open Addressing is a method for handling collisions. In Open Addressing, all elements are stored in the hash table itself. So at any point, the size of the table must be greater than or equal to the total number...

🕒 7 min read

Beam Penetration Technique in Computer Graphics

Pre-requisites: CRT, Refresh type output devices in Computer Graphics The cathode ray tube (CRT) monitor displays with the help of a combination of phosphorus, as phosphorus exists in multi-color. Bea...

🕒 2 min read

Collision Theory

Collision Theory says that when particles collide (strike) each other, a chemical reaction occurs. However, this is necessary but may not be a sufficient condition for the chemical reaction. The collision of molecu...

🕒 7 min read
